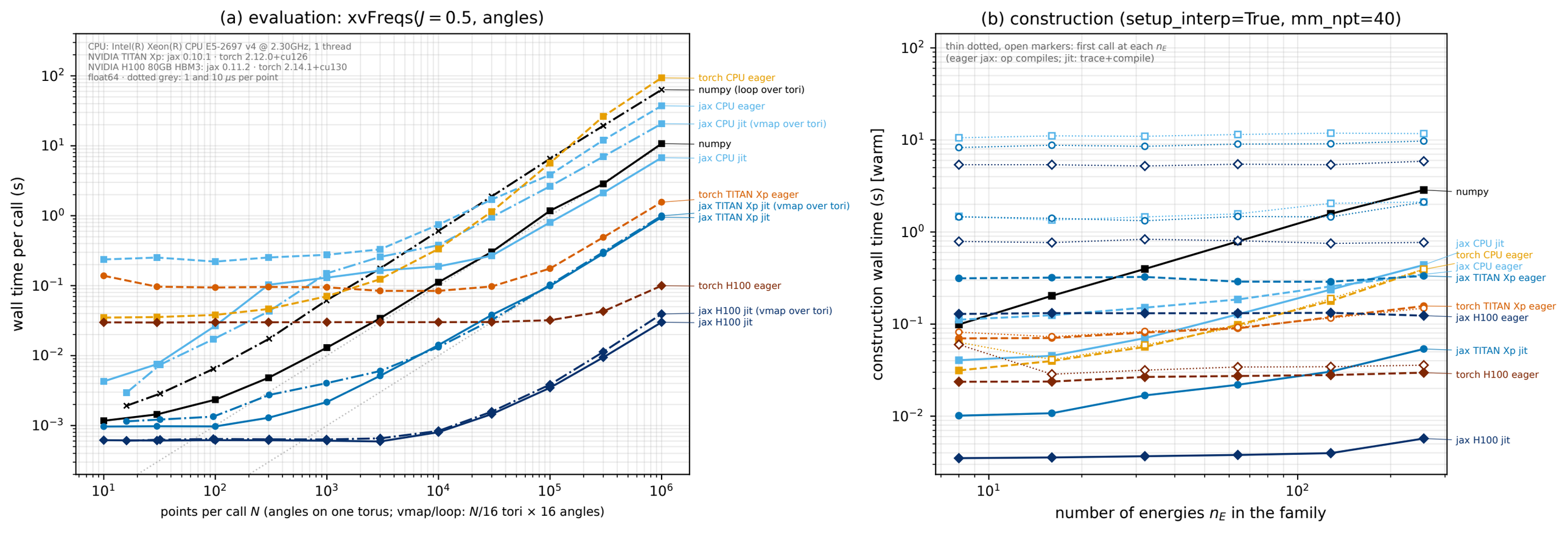
(a) evaluation: xvFreqs($J = 0.5$, angles)
CPU: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz, 1 thread
NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
NVIDIA H100 80GB HBM3: jax 0.11.2 · torch 2.14.1+cu130
float64 · dotted grey: 1 and 10 μs per point
wall time per call (s)
points per call N (angles on one torus; vmap/loop: N/16 tori × 16 angles)
torch CPU eager
numpy (loop over tori)
jax CPU eager
jax CPU jit (vmap over tori)
numpy
jax CPU jit
torch TITAN Xp eager
jax TITAN Xp jit (vmap over tori)
jax TITAN Xp jit
torch H100 eager
jax H100 jit (vmap over tori)
jax H100 jit
(b) construction (setup_interp=True, mm_npt=40)
thin dotted, open markers: first call at each n_E
(eager jax: op compiles; jit: trace+compile)
construction wall time (s) [warm]
number of energies n_E in the family
numpy
jax CPU jit
torch CPU eager
jax CPU eager
jax TITAN Xp eager
torch TITAN Xp eager
jax H100 eager
jax TITAN Xp jit
torch H100 eager
jax H100 jit